Using Large Language Models for Active Learning

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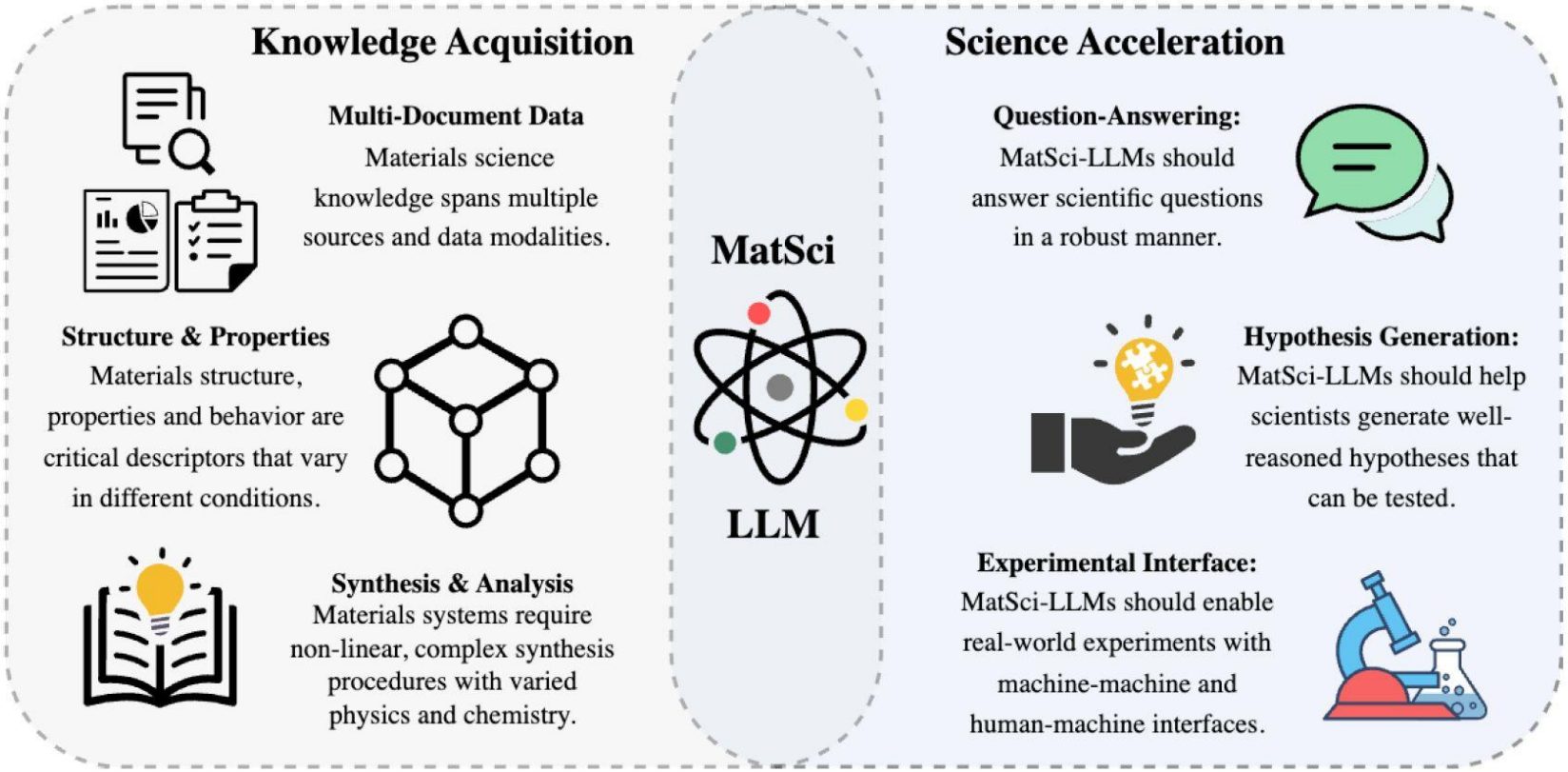
Large Language Models (LLMs) in Materials Research

- LLMs are AI models trained on vast amounts of text to understand and generate human-like language.

- They are powerful tools for
  - Domain-specific Q&A
  - Structure-property mapping
  - Analysis and proposal



➢ How can materials scientist leverage pre-trained LLMs to drive materials discovery

Miret, S., Krishnan, N.M.A., Nat Mach Intell 7, 991–998 (2025).

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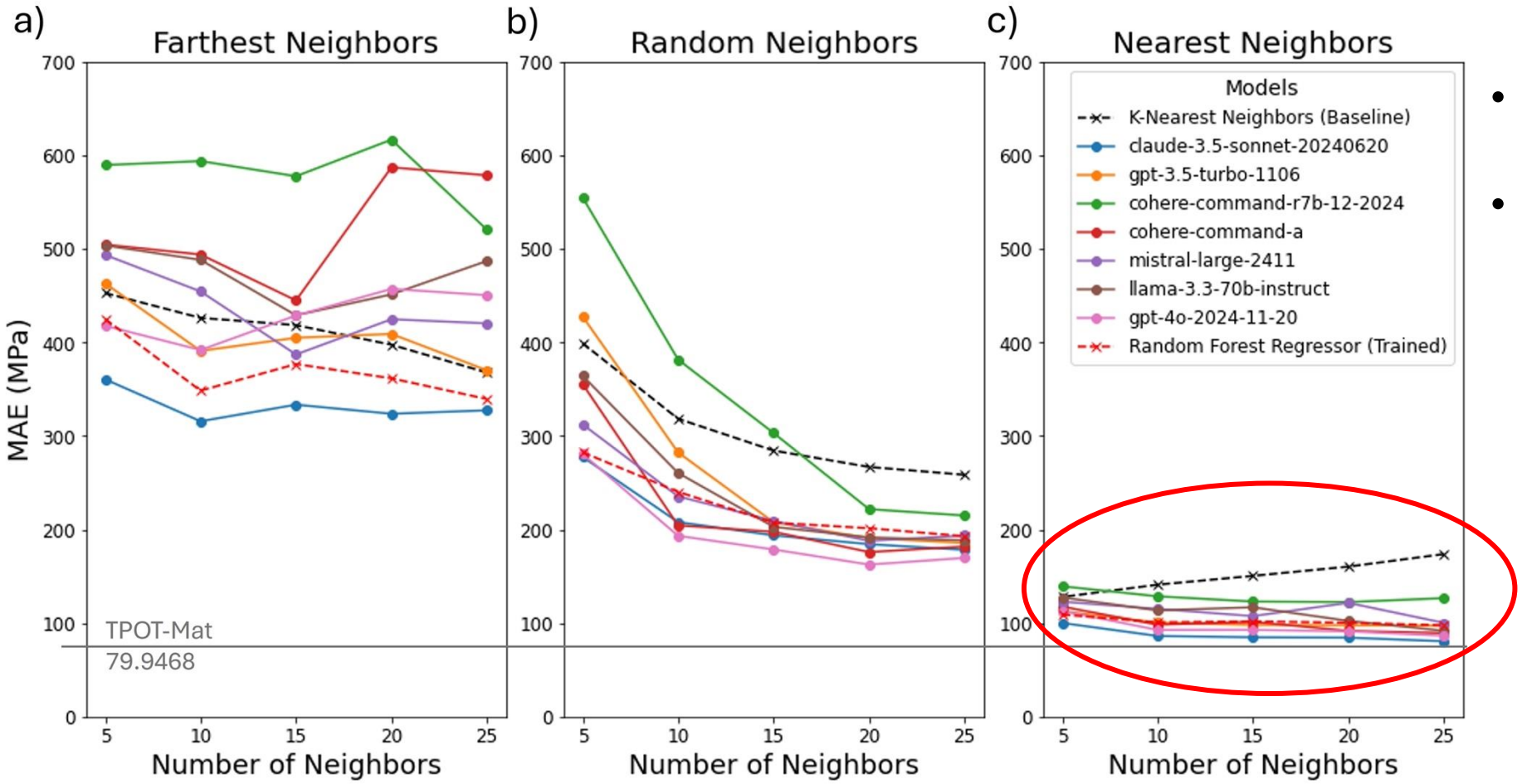
Using Pretrained LLMs for Materials Property Prediction



- Using **In-Context Learning** (ICL)
- **LLMs** can achieve **near-SOTA** performance in property predictions, when provided with **relevant data points** as few-shot examples

H. Wang et al., *Digital Discovery*, 2025, 4, 1612-1624

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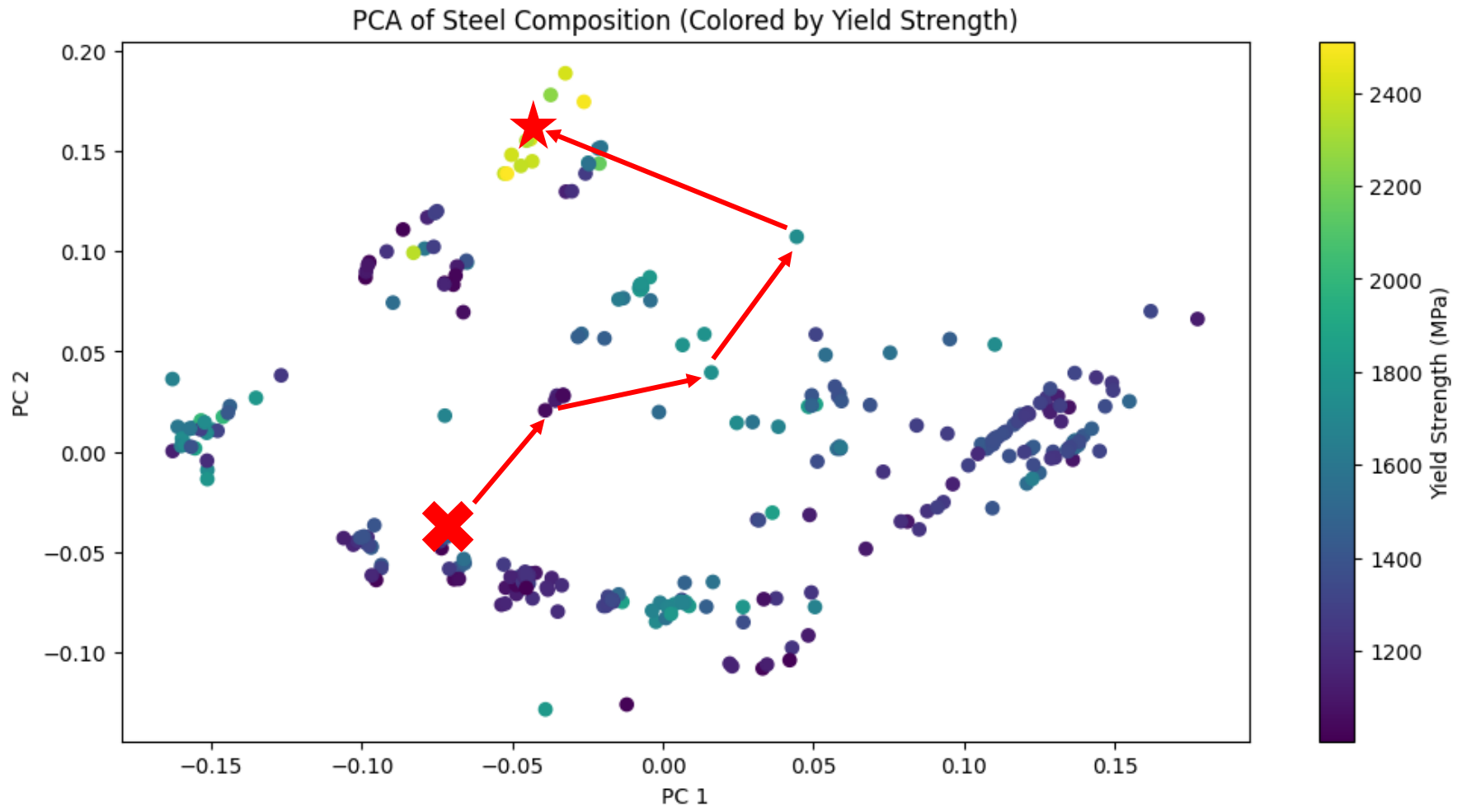
Using Pretrained LLMs for Materials Property Prediction



- What if letting LLM pick training points?
- Can LLMs iteratively suggest most informative points to find the best candidate?

-> Active Learning

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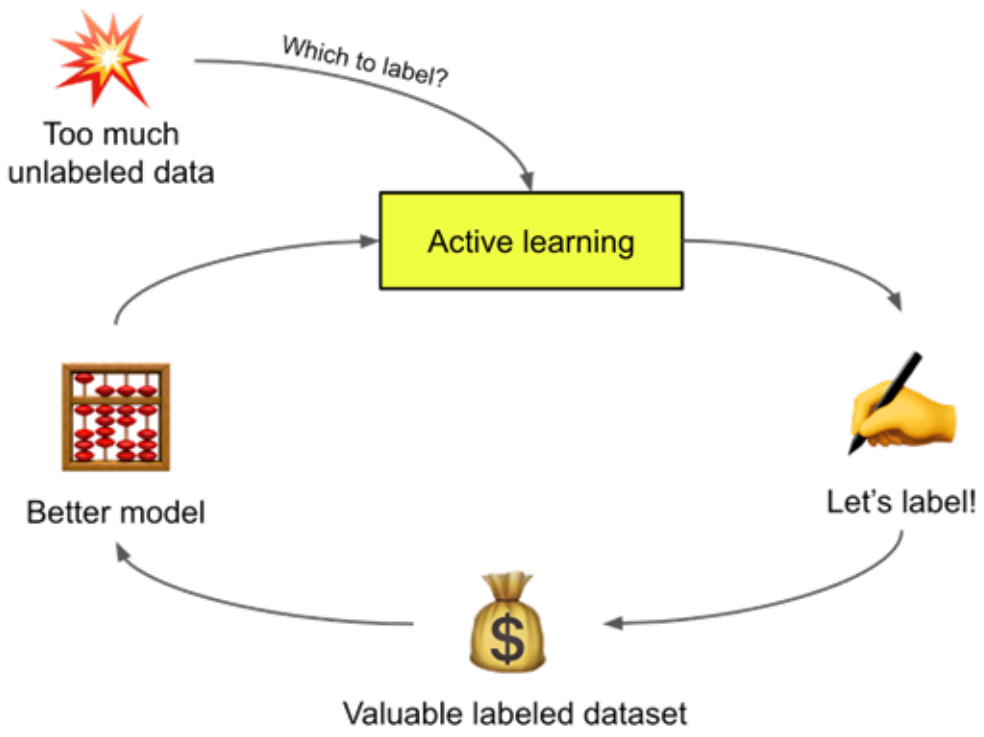
Active Learning



- Complex design space
- Start with few data
- More efficient than empirical methods and DoE

- **Traditional ML models**
  - **Cold-start problem**
    - Early rounds of experimentation have poor predictive performance and unreliable uncertainty estimation
  - **Hyperparameter tuning**
    - Model hyperparameters (e.g. learning rate, kernel values, exploration-exploitation trade-off value) are difficult to tune
  - **Generalizability**
    - Needs to train the model from scratch for a new problem
- **LLMs**
  - Prior knowledge from pre-training
  - No hyperparameter tuning
  - Shared feature space
  - Human-understandable feedbacks for iterative guidance

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Comparison of Workflows

	Traditional Active Learning	LLM-AL
Preprocessing	Data cleaning	
Training and Tuning	Featurization	**Prior knowledge**
	Acquisition Policy	**Prompt Engineering**
	Model tuning	**Prompt Engineering**
Selection	Prediction and Uncertainty	**Semantic Similarity**

Why is this relevant?

- Rapid deployment
- Not coding-intensive
- Useful and understandable experimental feedbacks

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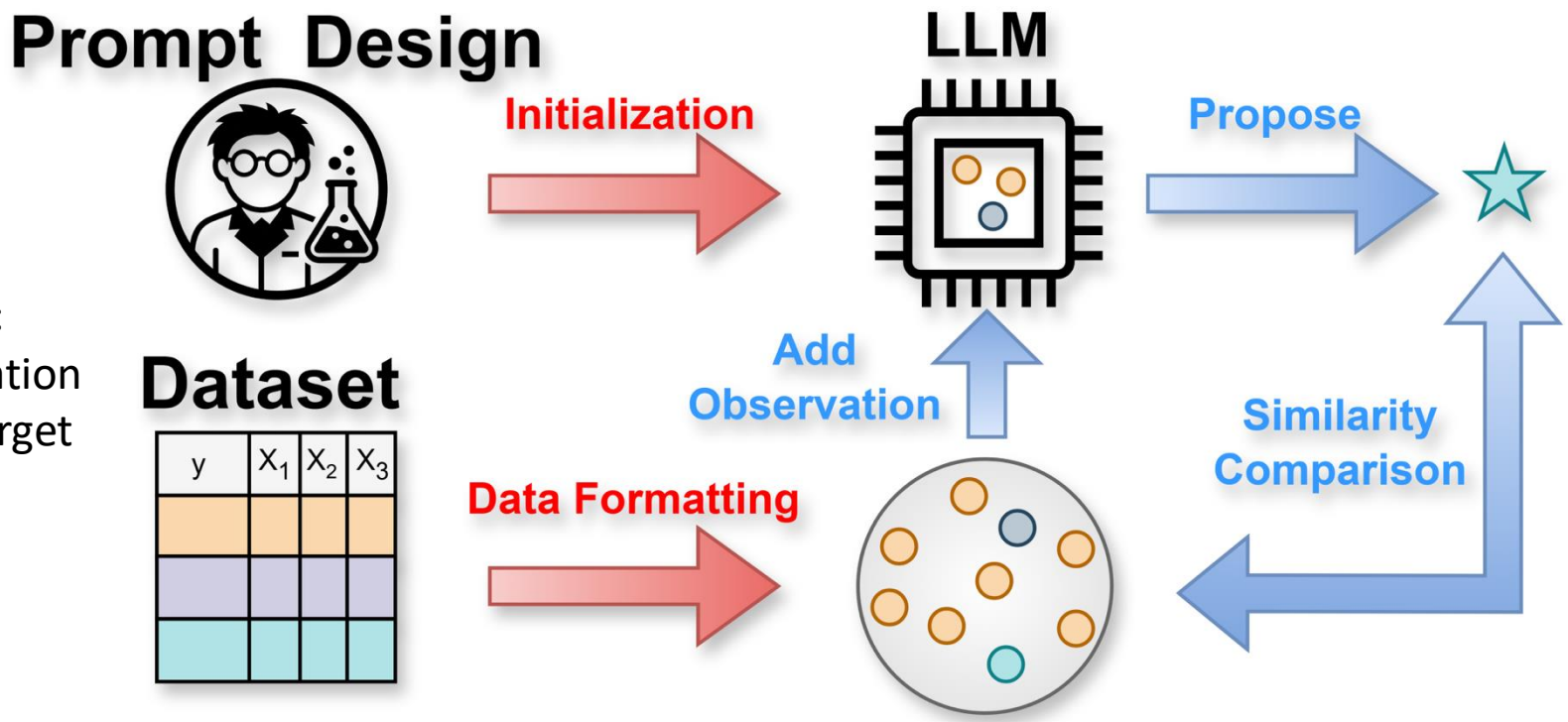
LLM-Based Active Learning (LLM-AL) Pipeline

Prompt Design:
1. Dataset-specific
2. Feature explanation
3. Optimization target



Active Learning Loop:
1. String slicing
2. Similarity comparison
3. Stopping criteria

Data Formatting:
1. Converting features into a string for each point
2. More complex description to give more contexts

H. Wang, R. E. Castañeda et al., *npj Comput Mater, under review*

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Notebook Time!

https://github.com/Toniaac/MRS2026Spring-AL-SDL-Tutorial/blob/main/LLM-AL/matbench_LLM-AL%20Pipeline.ipynb

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Benchmark Datasets

Dataset	Feature Dimension	Dataset Size	Optimization Target (y)	Goal	Data Source
matbench_steels	14	312	Yield Strength	Max.	Public
P3HT/CNT	5	323	Electrical Conductivity	Max.	Public
Perovskite	3	139	Instability Index	Min.	Public
Membrane	11	73	Elastic Modulus	Max.	Original

- Claude-sonnet-3.7, cohere-rerank-v3.5
- Two prompt strategies
- Benchmarking against GPR, RFR, XGBoost, BNN under UCB settings
- Performance and variability

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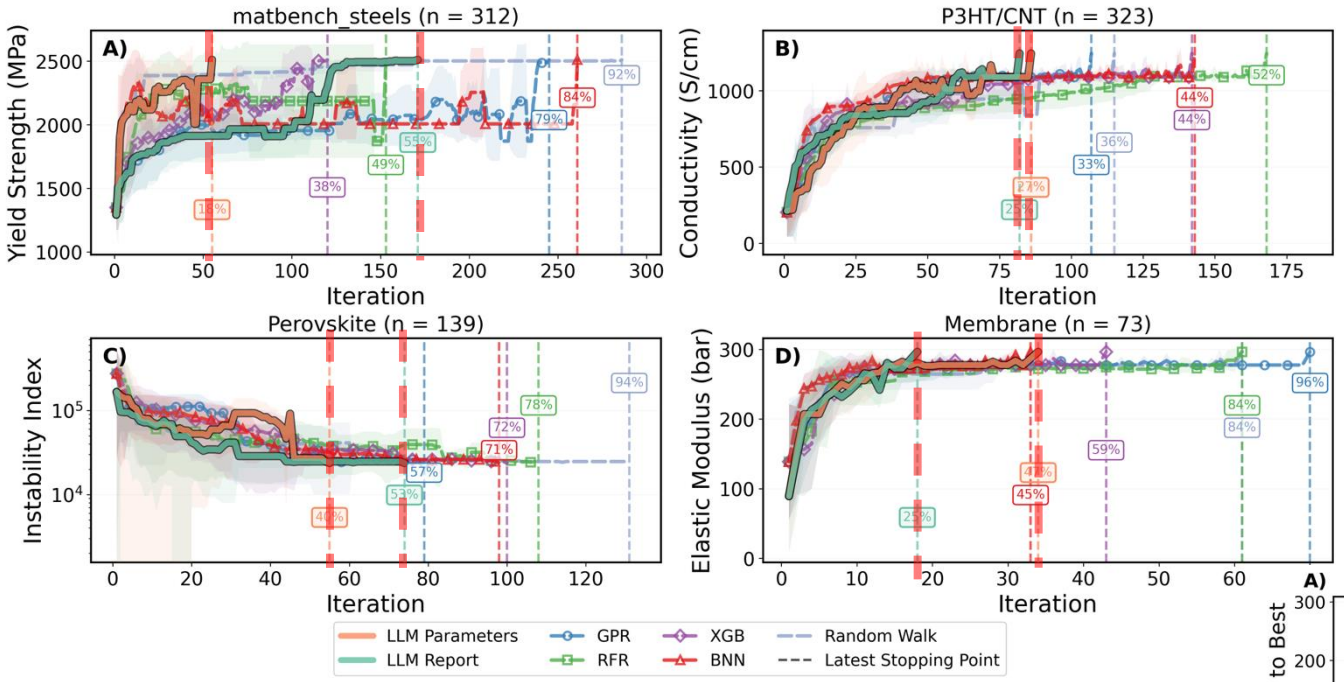
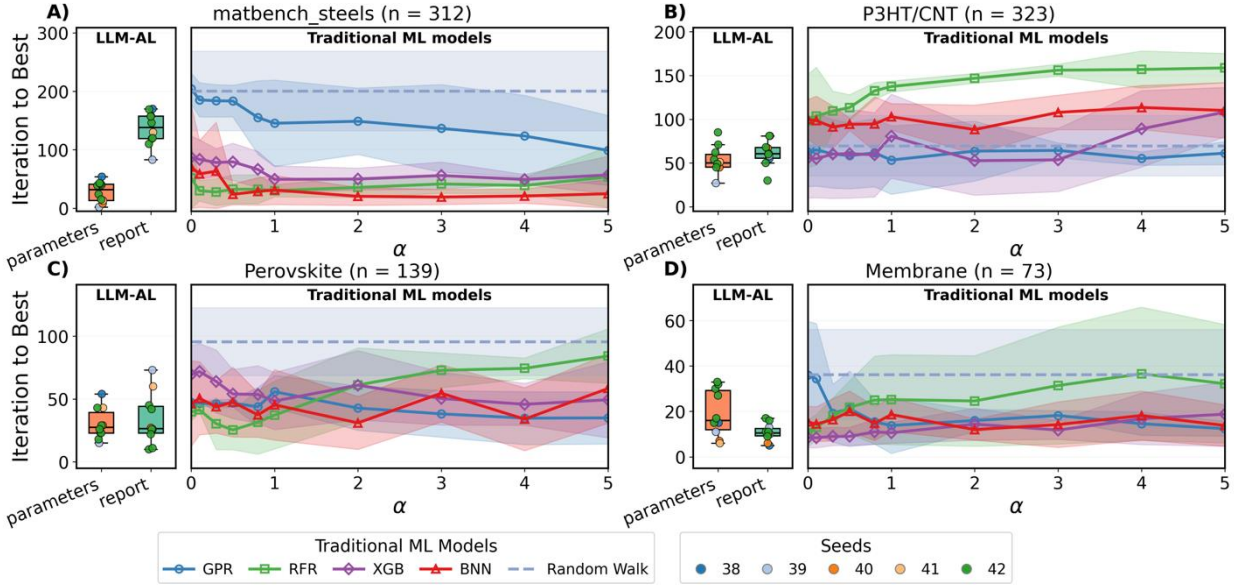
LLM-AL Performance Benchmarking



LLM-AL consistently outperforms traditional ML models

- **Matbench_steels**
  - Compositional & large parameter space
  - Benefit from concise prompts
- **Membrane Dataset**
  - Procedural & Experimental
  - Benefit from descriptive prompts



- Comparable or lower variability than traditional ML models
- Minimal hyperparameter tuning
- Minimal feature engineering

- **Non-determinism remains a challenge**
- **Not suitable for large input parameter size**

H. Wang, R. E. Castañeda et al., *npj Comput Mater, under review*

Conclusion

- Large language models show strong potential in materials research
- LLM-based active learning reduces the need for heavy feature engineering and hyperparameter tuning.
- Pretrained LLMs can outperform conventional machine learning in low-data and cold-start active learning settings.
- LLMs are promising tools for autonomous, data-efficient, and interpretable materials discovery.

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Acknowledgement

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